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Questions

Q1.

The oxidation number of sulfur in the sulfate ion, SO_4^{2-}, is

- ☐ **A** −2
- ☐ **B** +4
- ☐ **C** +6
- ☐ **D** +8

Use this space for any rough working. Anything you write in this space will gain no credit.

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(Total for question = 1 mark)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Dichromate(VI) ions may be reduced in acidic solution.

$$Cr_2O_7^{2-}(aq) + xH^+(aq) + ye^- \rightarrow 2Cr^{3+}(aq) + zH_2O(l)$$

The coefficients in this half-equation are

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	x	y	z
☐ **A**	14	6	7
☐ **B**	14	3	7
☐ **C**	7	6	3.5
☐ **D**	7	3	3.5

(Total for question = 1 mark)

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Q3.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Iodate(V) ions, IO_3^-, oxidise dithionate ions, $S_2O_6^{2-}$, according to the equation

$$\mathbf{x}\ IO_3^- + \mathbf{y}\ S_2O_6^{2-} + 4H_2O \rightarrow \mathbf{z}\ SO_4^{2-} + 8H^+ + I_2$$

What are the balancing numbers **x**, **y** and **z**?

		x	y	z
☐	**A**	2	1	2
☐	**B**	2	2	4
☐	**C**	2	5	5
☐	**D**	2	5	10

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(Total for question = 1 mark)

Q4.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about the reaction shown.

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$$2KMnO_4 + xH_2C_2O_4 + yH_2SO_4 \rightarrow 2MnSO_4 + K_2SO_4 + 10CO_2 + zH_2O$$

(a) What values of x, y and z are needed to balance the equation?

(1)

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	x	y	z
☐ A	5	6	8
☐ B	10	3	4
☐ C	5	3	8
☐ D	10	6	4

(b) What is the reducing agent in the reaction?

(1)

☐ **A** H^+

☐ **B** $C_2O_4^{2-}$

☐ **C** MnO_4^-

☐ **D** SO_4^{2-}

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(Total for question = 2 marks)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A student is provided with 25.0 cm^3 of 1.00 mol dm^{-3} hydrochloric acid.

What volume of distilled water should the student add to this solution to make a 0.0500 mol dm^{-3} solution?

☐ **A** 25.0 cm^3

☐ **B** 50.0 cm^3

☐ **C** 475 cm^3

☐ **D** 500 cm^3

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(Total for question = 1 mark)

Q6.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Magnesium reacts with hydrochloric acid.

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$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

Which statement about this reaction is correct?

☒ **A** magnesium atoms act as oxidising agents

☒ **B** hydrogen molecules act as reducing agents

☒ **C** hydrogen ions act as oxidising agents

☒ **D** chloride ions act as oxidising agents

(Total for question = 1 mark)

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Q7.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the oxidation number of chromium in $Na_2Cr_2O_7$?

☒ **A** +1
☒ **B** +2
☒ **C** +3
☒ **D** +6

(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Hydrogen peroxide, H_2O_2 , breaks down into water and oxygen.

In terms of oxidation and reduction, how do hydrogen and oxygen change in this reaction?

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	Hydrogen	Oxygen
☒ **A**	oxidised	reduced
☒ **B**	oxidised and reduced	unchanged
☒ **C**	reduced	oxidised
☒ **D**	unchanged	oxidised and reduced

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(Total for question = 1 mark)

Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the oxidation number of phosphorus in the phosphate ion, PO_4^{3-}

☒ **A** –3
☒ **B** +3
☒ **C** +5
☒ **D** +7

(Total for question = 1 mark)

Q10.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In an oxide of nitrogen, the oxidation number of nitrogen is +4.

Which is the formula of the oxide?

☒ **A** N_2O
☒ **B** N_2O_3
☒ **C** N_2O_4
☒ **D** N_2O_5

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(Total for question = 1 mark)

Q11.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

This question is about the reaction of potassium iodide with concentrated sulfuric acid.

(a) Which of these would **not** be seen?

(1)

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- ☐ **A** misty fumes
- ☐ **B** black solid
- ☐ **C** yellow solid
- ☐ **D** dense white smoke

(b) One of the reaction products is the gas H_2S.

What is the **change** in oxidation number of sulfur as H_2S forms?

(1)

- ☐ **A** −8
- ☐ **B** −6
- ☐ **C** −2
- ☐ **D** +6

(Total for question = 2 marks)

Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which reaction is **not** a redox reaction?

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- ☐ **A** $4KClO_3(s) \rightarrow 3KClO_4(s) + KCl(s)$
- ☐ **B** $2HCl(aq) + Ba(OH)_2(aq) \rightarrow BaCl_2(aq) + 2H_2O(l)$
- ☐ **C** $Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$
- ☐ **D** $Cl_2(g) + H_2O(l) \rightarrow HCl(aq) + HClO(aq)$

(Total for question = 1 mark)

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Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How does an oxidising agent change during a redox reaction?

		Electrons	Oxidation number
☐	**A**	gains electrons	decreases
☐	**B**	gains electrons	increases
☐	**C**	loses electrons	decreases
☐	**D**	loses electrons	increases

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(Total for question = 1 mark)

Q14.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Two half-equations for a reaction are shown.

$$Cu \rightarrow Cu^{2+} + 2e^-$$

$$NO_3^- + 4H^+ + 3e^- \rightarrow NO + 2H_2O$$

What is the overall ionic equation for this reaction?

☐ **A** $Cu + NO_3^- + 4H^+ \rightarrow Cu^{2+} + NO + 2H_2O$

☐ **B** $2Cu + NO_3^- + 4H^+ \rightarrow 2Cu^{2+} + NO + 2H_2O$

☐ **C** $3Cu + 2NO_3^- + 8H^+ \rightarrow 3Cu^{2+} + 2NO + 4H_2O$

☐ **D** $6Cu + 2NO_3^- + 8H^+ \rightarrow 6Cu^{2+} + 2NO + 4H_2O$

(Total for question = 1 mark)

Q15.

For each question, select one answer from A to D and put a cross in the box ☒. If you change

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your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Barium hydroxide reacts with sulfuric acid as shown.

$$Ba(OH)_2(aq) + H_2SO_4(aq) \rightarrow BaSO_4(s) + 2H_2O(l)$$

Which is the ionic equation for this reaction?

☐ **A** $Ba^{2+}(aq) + 2OH^-(aq) + 2H^+(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s) + 2H_2O(l)$

☐ **B** $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$

☐ **C** $OH^-(aq) + H^+(aq) \rightarrow H_2O(l)$

☐ **D** $Ba^{2+}(aq) + 2OH^-(aq) + 2H^+(aq) + SO_4^{2-}(aq) \rightarrow Ba^{2+}(s) + SO_4^{2-}(s) + 2H_2O(l)$

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(Total for question = 1 mark)

Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the name of the compound with the formula $KClO_2$?

☐ **A** potassium chlorate(II)
☐ **B** potassium chlorate(III)
☐ **C** potassium chlorate(IV)
☐ **D** potassium chlorate(V)

(Total for question = 1 mark)

Q17.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the oxidation number of bromine in BrO_3^-?

☐ **A** +1
☐ **B** +3
☐ **C** +5
☐ **D** +7

(Total for question = 1 mark)

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Q18.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In which compound is the oxidation number of nitrogen +5?

- ☐ **A** $Ca(NO_3)_2$
- ☐ **B** Mg_3N_2
- ☐ **C** N_2O_3
- ☐ **D** $NaNO_2$

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(Total for question = 1 mark)

Q19.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In which reaction is the copper species acting as an oxidising agent?

- ☐ **A** $Cu^{2+} + 2Ag \rightarrow 2Ag^{+} + Cu$
- ☐ **B** $2Cu^{+} + O^{2-} \rightarrow Cu_2O$
- ☐ **C** $3Cu + O_2 \rightarrow Cu_2O + CuO$
- ☐ **D** $Cu + Hg^{2+} \rightarrow Hg + Cu^{2+}$

(Total for question = 1 mark)

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Q20.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which name is correct for the ion SO_4^{2-} ?

- ☐ **A** sulfate(II)
- ☐ **B** sulfate(IV)
- ☐ **C** sulfate(VI)
- ☐ **D** sulfate(VIII)

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(Total for question = 1 mark)

Q21.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

What is the value of n in the half-equation shown?

$$O_2(g) + 2H_2O(l) + ne^- \rightarrow nOH^-(aq)$$

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- ☐ **A** 2
- ☐ **B** 3
- ☐ **C** 4
- ☐ **D** 5

(Total for question = 1 mark)

Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the formula of potassium manganate(VI)?

- ☐ **A** $KMnO_4$
- ☐ **B** K_2MnO_4
- ☐ **C** K_3MnO_4
- ☐ **D** $K_6Mn_2O_6$

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(Total for question = 1 mark)

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Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In which pair of compounds is the stated element in the **same** oxidation state?

- ☐ **A** chlorine in Cl_2O_7 and $Ca(ClO_3)_2$
- ☐ **B** chromium in $K_2Cr_2O_7$ and K_2CrO_4
- ☐ **C** manganese in K_2MnO_4 and Mn_2O_7
- ☐ **D** oxygen in K_2O and K_2O_2

(Total for question = 1 mark)

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Q24.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which ion contains vanadium with an oxidation number of **+4**?

- ☐ **A** VO^{2+}
- ☐ **B** VO_2^+
- ☐ **C** VO_3^-
- ☐ **D** VO_4^{3-}

(Total for question = 1 mark)

Q25.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

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In a neutralisation reaction, 20.0 cm^3 of 0.500 mol dm^{-3} nitric acid reacts with 10.0 cm^3 of 1.00 mol dm^{-3} aqueous sodium hydroxide.

$$HNO_3(aq) + NaOH(aq) \rightarrow NaNO_3(aq) + H_2O(l)$$

What is the concentration, in mol dm^{-3}, of the sodium nitrate solution produced?

- ☐ **A** 0.33
- ☐ **B** 0.50
- ☐ **C** 0.67
- ☐ **D** 1.0

(Total for question = 1 mark)

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Q26.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the formula of barium(II) ferrate (VI)?

- ☒ **A** Ba_2FeO_3
- ☒ **B** $BaFeO_3$
- ☒ **C** Ba_2FeO_4
- ☒ **D** $BaFeO_4$

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(Total for question = 1 mark)

Q27.

Calcium hypochlorite, $Ca(ClO)_2$, is used for water treatment in swimming pools. It is produced in the reaction between $Ca(OH)_2$ and Cl_2.

$$2Ca(OH)_2(aq) + 2Cl_2(g) \rightarrow Ca(ClO)_2(aq) + CaCl_2(aq) + 2H_2O(l)$$

(a) State the type of reaction occurring in the production of $Ca(ClO)_2$. Justify your answer using oxidation numbers.

(3)

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(b) Calculate the percentage atom economy by mass for the production of $Ca(ClO)_2$ in this reaction.

(2)

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(c) A swimming pool has the dimensions 50 m × 25 m × 2.0 m.

The water in this swimming pool has a $Ca(ClO)_2$ concentration of 4.2 mg dm^{-3} .

(i) Calculate the mass, **in kg**, of $Ca(ClO)_2$ required to treat the water needed to completely fill this swimming pool.

(3)

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(ii) Calculate the volume of Cl_2, at room temperature and pressure, needed to make the mass of $Ca(ClO)_2$ calculated in (c)(i).

(3)

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(Total for question = 11 marks)

Q28.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which equation shows a disproportionation reaction?

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☐ **A** $Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2H_2O + 2NO_2$

☐ **B** $I_2 + 5O_3 + H_2O \rightarrow 2HIO_3 + 5O_2$

☐ **C** $2MnO_4^- + MnO_2 + 4OH^- \rightarrow 3MnO_4^{2-} + 2H_2O$

☐ **D** $S_2O_3^{2-} + 2H^+ \rightarrow SO_2 + S + H_2O$

(Total for question = 1 mark)

Q29.

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Zinc metal reacts with copper(II) sulfate solution. The equation for the reaction is:

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s) \qquad \Delta_r H = -210 \text{ kJ mol}^{-1}.$$

(a) What is the temperature rise, in °C, when excess zinc powder is added to 50 cm^3 of copper(II) sulfate solution containing 0.0025 mol of copper(II) ions?

[Assume the specific heat capacity of the solution is 4.2 J g^{-1} $°C^{-1}$].

(1)

☐ **A** 2.5
☐ **B** 10.5
☐ **C** 25.0
☐ **D** 44.1

(b) The reaction of zinc with copper(II) sulfate is best classified as:

(1)

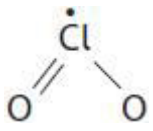
☐ **A** disproportionation
☐ **B** neutralisation
☐ **C** redox
☐ **D** thermal decomposition

(Total for question = 2 marks)

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Q30.

Chlorine dioxide, ClO_2, can be used to treat water to make it safe for drinking. The molecule is a free radical. One suggested structure is shown.



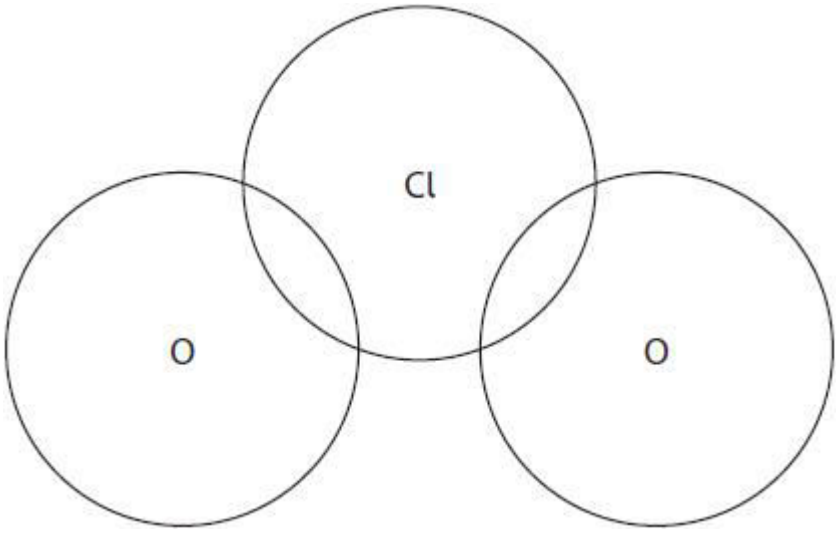
(a) (i) Complete the dot-and-cross diagram for this structure.

Use dots (•) for oxygen electrons and crosses (×) for chlorine electrons.

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(2)



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(ii) Give the reason why chlorine dioxide is not a linear molecule.

(1)

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(b) Chlorine dioxide can be made from potassium chlorate(V), $KClO_3$, in a two-step process.

Step **1** Potassium chlorate(V) reacts with concentrated sulfuric acid to form compound **A**.
Step **2** Compound **A** then undergoes a disproportionation reaction to form chlorine dioxide and hydrogen chlorate(VII), $HClO_4$.

(i) State the meaning of the Roman numerals in chlorate(V) and chlorate(VII).

(1)

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.....

(ii) State the oxidation number of chlorine in chlorine dioxide.

(1)

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(iii) In Step **2**, compound **A** undergoes disproportionation.
Deduce the possible range of oxidation numbers for the chlorine in compound **A**. Justify your answer.

(2)

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(iv) Compound **A** has M_r = 84.5, and a composition by mass of H = 1.18 %, Cl = 42.01 % and the remainder is oxygen.

Deduce the molecular formula of compound **A**, showing **all** your working.

(4)

(v) Write an equation for each of the two steps in the formation of chlorine dioxide, using your answers in (b).

State symbols are not required.

(2)

(Total for question = 13 marks)

Q31.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which compound does **not** contain an element with an oxidation number of −1?

- ☒ **A** HF
- ☒ **B** NaH
- ☒ **C** H_2O_2
- ☒ **D** CH_4

(Total for question = 1 mark)

Q32.

Answer the question with a cross in the box you think is correct (☒). If you change your mind

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about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Sulfuric acid and sodium hydroxide react as shown.

$$H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

Which sample of sulfuric acid solution will be exactly neutralised by 20 cm^3 of 0.5 mol dm^{-3} sodium hydroxide?

	Volume of sulfuric acid / cm^3	Concentration of sulfuric acid / mol dm^{-3}
☐ **A**	20	0.25
☐ **B**	20	0.5
☐ **C**	40	0.5
☐ **D**	10	1.0

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(Total for question = 1 mark)

Q33.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which reaction involves the oxidation of a Group 7 element?

☐ **A** $Br_2 + F_2 \rightarrow 2BrF$

☐ **B** $U + 3ClF_3 \rightarrow UF_6 + 3ClF$

☐ **C** $2F_2 + 2H_2O \rightarrow 4HF + O_2$

☐ **D** $KCl + H_2SO_4 \rightarrow HCl + KHSO_4$

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(Total for question = 1 mark)

Q34.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Bromine, $Br_2(l)$, can be disposed of safely by reaction with aqueous sodium thiosulfate, $Na_2S_2O_3(aq)$.

$$Br_2(l) + 2Na_2S_2O_3(aq) \rightarrow 2NaBr(aq) + Na_2S_4O_6(aq)$$

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What is the minimum volume, in cm^3, of 0.5 mol dm^{-3} $Na_2S_2O_3(aq)$ needed to react completely with 0.01 mol of $Br_2(l)$?

- **A** 0.01
- **B** 0.04
- **C** 10
- **D** 40

(Total for question = 1 mark)

Q35.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the mass of nitrogen **atoms** in 50 cm^3 of a 2.0 mol dm^{-3} solution of ammonium nitrate, $NH_4NO_3(aq)$?

- **A** 35 g
- **B** 8.0 g
- **C** 2.8 g
- **D** 1.4 g

(Total for question = 1 mark)

Q36.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Equal volumes of some 0.01 mol dm^{-3} solutions were mixed.

Which pair of solutions gave the greatest mass of precipitate?

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- **A** $BaCl_2$ and $AgNO_3$
- **B** $BaCl_2$ and Ag_2SO_4
- **C** $BaCl_2$ and $NaNO_3$
- **D** $BaCl_2$ and Na_2SO_4

(Total for question = 1 mark)

Q37.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is the correct order of **increasing** oxidation number of nitrogen in NO, NO_3^- and NH_3?

- ☐ **A** NH_3 NO NO_3^-
- ☐ **B** NH_3 NO_3^- NO
- ☐ **C** NO NH_3 NO_3^-
- ☐ **D** NO_3^- NH_3 NO

(Total for question = 1 mark)

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Q38.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The most common sources of iodine contain the iodate ion IO_3^-.
Iodate ions react with iodide ions in acidic conditions as shown.

$$5I^- + IO_3^- + 6H^+ \rightarrow 3I_2 + 3H_2O$$

The forward reaction is best described as

- ☐ **A** disproportionation
- ☐ **B** neutralisation
- ☐ **C** oxidation
- ☐ **D** redox

(Total for question = 1 mark)

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